THE STORY OF ARMS AND ARMOR

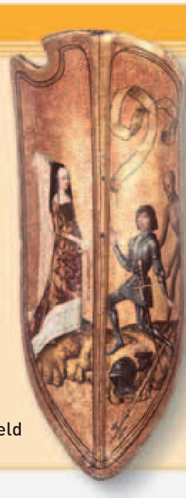
FROM STICKS AND STONES TO A DEADLY ART FORM

Whether for hunting or sport, conflict or contests of skill, handheld arms have played a crucial role in human existence and advancement. The first weapons developed out of survival tools: found objects, such as stones, were used to bludgeon prey, or to fend off predatory animals or rival humans.

As prehistoric humans' skills advanced, clubs and stone hand-axes gave way to carefully crafted wooden spears used to hunt animals or impale fish. Even more effective weapons married wooden shafts with razor-sharp flint blades to form axes, daggers, spears, and arrowheads. Soft, easily worked metals such as copper replaced flint, followed by stronger, sharper, and longer Bronze Age and Iron Age swords, daggers, javelins, and battle-axes. Until the advent of firearms, the history of handheld weapons is one of variations on a theme, culminating in the sophisticated forging processes of Japan's samurai swords, which at their height in the 14th–16th centuries wrapped super-sharp steel around a flexible iron core.

SHIELDS

Like arms and body armor, shields—a type of "accessory armor"—could be functional, decorative, or both. During the medieval period in Europe, when knights held high status in society, shields were often embellished with elaborate scenes of courtly devotion or prowess in battle. Decoration like this was thought to bring added protection to the bearer.



15th-century Flemish shield

THE DEVELOPMENT OF ARMOR

Early "armor" consisted of padding: thick layers of cloth with a stiff leather "helmet" to protect the head. Plated helmets, breastplates, and wooden shields were used by classical Greek and Roman armies, but elsewhere, ordinary soldiers relied on padding, leather, and luck—a situation that changed little in Europe until chain mail was perfected in 11th-century France. Full suits of armor were costly, so they were also used as status symbols.

French 16th-century embossed helmet
Armor reached its greatest decorative heights during the Renaissance. Suits and helmets were embossed and etched, gilded or silvered, particularly for tournaments—and to show the owner's wealth and status.

750,000–50,000 BCE
Flint cutting edges
Razor-sharp flint daggers, spears, and axes are used for both hunting and warfare.

Flint dagger

450,000–400,000 BCE
Wooden weapons
Easily worked and readily available, wood is shaped into spears for hunting or defense.

Wooden spear

5500–3300 BCE
Flint arrowheads
The wooden bow combined with arrowheads made from sharpened flint proves a deadly combination, allowing users to strike their victims from a safe distance.

3700–2300 BCE
Metal weapons
Metalworking gives rise to sophisticated and effective blades in the Bronze and Iron Ages.

Bronze ax

2500 BCE
Helmets
The first part of the body to be protected is the head. Early armies use plated helmets, but most soldiers rely on leather caps.

Attic helmet

c. 1400 BCE
Suit armor develops
Plated body armor is an early invention, but it is expensive and not always practical for movement in battle.

Mycenaean armor

6th–mid-5th centuries BCE
The crossbow
Crossbows can be cocked well in advance of firing—providing one of the earliest "loaded" weapons.

Early Chinese crossbow

3rd–4th centuries
Steel blades
Adding carbon to iron produces steel, which allows bladed weapons to be mass-produced. Blades also become stronger and longer.

Roman gladius